



Impacts of shellfish and macroalgae mariculture on the seawater carbonate system and air-sea CO₂ flux in Haizhou Bay, China

Jia-Hui Zhou^a, Ying-Zi Guo^a, Xu Zhao^a, Jin-Chen Yu^{b,**}, Jin Zhou^b, Jun Lin^a, Hong Huang^{a,*}

^a College of Oceanography and Ecological Science, Shanghai Ocean University, 999 Hucheng Ring Road, Shanghai, 201306, China

^b East China Sea Fisheries Research Institute, Chinese Academy of Fishery Science, 300 Jungong Road, Shanghai, 200090, China

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ABSTRACT

China is the largest mariculture country, and shellfish and algae output ranks first, showing high carbon sink capacity. In recent years, the single cultivation of macroalgae (*Pyropia yezoensis*) has been changed to macroalgae-shellfish mariculture in Haizhou Bay to increase the yield of *P. yezoensis* and improve the water environment quality. In this study, four surveys were carried out in July 2022 during the monoculture period of oyster (*Magallana gigas*), as well as at different stages of *P. yezoensis* culture (head-crop period, November 2022, peak growing season, January 2023, and end of harvesting, March 2023) in the mariculture and the surrounding waters of Haizhou Bay. The effects of different stages of culture on the seawater environment and seasonal and spatial variations in the carbonate system were analyzed, and the carbon sink capacity was preliminarily estimated. The results showed that in summer, the calcification of *M. gigas* and the primary production process of phytoplankton effectively reduced the dissolved inorganic carbon (DIC) level in the culture area. The culture area acts as a CO₂ sink, with an average air-sea CO₂ flux of $-4.5 \text{ mmol m}^{-2} \text{ d}^{-1}$. During the polyculture period, the *P. yezoensis* culture activities maintained the stability of the seawater carbonate system, and the culture area shows strong CO₂ sinks, with the average air-sea CO₂ flux of $-24.10 \text{ mmol m}^{-2} \text{ d}^{-1}$, $-37.68 \text{ mmol m}^{-2} \text{ d}^{-1}$, and $-38.99 \text{ mmol m}^{-2} \text{ d}^{-1}$, respectively. The absorption of CO₂ by large-scale cultured *P. yezoensis* through the "biological pump" effect is the main factor affecting the CO₂ exchange process at the air-sea interface, and the absorption rate of CO₂ by *P. yezoensis* at the mature stage is higher than that at the growth stage before harvesting. The study revealed that macroalgae-shellfish mariculture could promote mutual growth, alleviate environmental pressure, and enhance the carbon sink of the culture area. The relationship between mariculture and the carbon cycle of a mariculture ecosystem is very complicated, and its biochemical process should be given great attention for further study.

1. Introduction

In the context of global warming, the carbon sequestration capacity of marine ecosystems has been increasingly emphasized as the ocean, the largest carbon reservoir other than the lithosphere, absorbs about one-third of anthropogenic carbon dioxide (CO₂) emissions (Sabine et al., 2004; Yool and Fasham, 2001). Currently, through the scientific grasp of the ocean carbon cycle process and mechanism, marine biological carbon sequestration is an effective way to increase the ocean carbon sink (Tang et al., 2011). Mariculture is an essential human activity in coastal areas, where farmed organisms affect the carbon cycle of coastal ecosystems by fixing and converting CO₂ (Pan et al., 2019).

Macroalgae are primary producers in coastal ecosystems and they are highly productive through photosynthesis and nutrient support (Gao et al., 2021; Ikawa and Oechel, 2015). Compared to other photosynthetic autotrophic organisms in the ocean, macroalgae play a significant role in mitigating climate change through extensive nearshore mariculture. The sufficient expansion space and longer growth cycle, as well as the assistance of mariculture facilities such as floating rafts, significantly enhance the carbon sequestration capacity of macroalgae (Gao et al., 2022; Yan et al., 2011). Shellfish are an essential group of nearshore ecosystems, and their use of dissolved inorganic carbon (DIC) in seawater to form shells in the calcification process can significantly affect the carbonate system of the surrounding seawater, especially in

* Corresponding author.

** Corresponding author.

E-mail addresses: dhuyjincheng@163.com (J.-C. Yu), hhuang@shou.edu.cn (H. Huang).

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the nearshore waters where the seawater exchange rate is low (Frankignoulle et al., 1994; Li et al., 2021). Moreover, the biological sediment deposition formed by shellfish metabolism can accelerate the vertical transport of carbon and promote the formation of sedimentary carbon pools (Giles and Pilditch, 2004; Jiang et al., 2015; Zhang et al., 2017).

In an ideal integrated algae-shellfish polyculture system, the nitrogen and phosphorus released by shellfish can alleviate the problem of excessive consumption of nutrients in seawater during algae growth. In contrast, the organic debris and oxygen (O_2) produced by algae ensure the various metabolic activities of shellfish (Fang et al., 2020; Xiao et al., 2021). Bartoli et al. (2001) showed that the regeneration of nutrients in nearshore waters can increase the primary productivity of macroalgae. Macroalgae and shellfish can promote each other's growth, reduce environmental stress, and enhance the carbon sink function of mariculture systems. Taking Daya Bay as an example, Han et al. (2017) found that the *M. gigas* cultivation area in Daya Bay exhibits a source of atmospheric CO_2 , and in the comprehensive cultivation area of *M. gigas* and *Gracilaria*, the cultivation of shellfish and algae effectively reduces the release of CO_2 .

Comprehensive studies have been conducted on the air-sea CO_2 flux and carbonate systems in open waters (Qu et al., 2015; Shadwick et al., 2011; Wu et al., 2024). However, research on carbonate systems in nearshore culture areas mainly focuses on the mature integrated mariculture model of Sanggou Bay, analyzing the contribution of organisms to the carbon cycle in Sanggou Bay under different culture models (Liu et al., 2022; Zhang et al., 2013). Haizhou Bay is a typical mariculture bay in China, with the production of *P. yezoensis* accounting for 1/4 of the country's total. However, currently, there are few reports on the impact of large-scale mariculture activities in Haizhou Bay on carbonate systems, especially on the interactions between organisms in different culture stages. The primary purpose of this study is to (1) attempt to identify the impact of mariculture activities on the water environment and carbonate systems, as well as the interactions between culture species; (2) Evaluate the carbon sequestration function of the study area based on the cultivation cycle of *P. yezoensis* and the seasonal characteristics of cultivation activities; The results of this study contribute to filling the knowledge gap in the regulation mechanism of the carbonate system in Haizhou Bay by *P. yezoensis*-*M. gigas* polyculture, and can also provide scientific basis for local mariculture development planning.

2. Material and methods

2.1. Study area

Haizhou Bay is a semi-open bay in the western part of the South Yellow Sea of China. Affected by the topography, the nearshore tidal current presents a reciprocating current. It has a temperate monsoon climate with mild temperatures and plenty of sunshine. Also, injecting several freshwater rivers brings rich nutrients to the bay (Zhou et al., 2024). The unique conditions provide a good external environment for the propagation of *P. yezoensis*. However, in recent years, due to the irrational culture density, the frequent occurrence of *P. yezoensis* diseases has led to a decline in yield (Gong, 2022). In order to make rational use of the culture area, Haizhou Bay has turned to the polyculture mode of *P. yezoensis*-*M. gigas* to replace the traditional monoculture model. *Pyropia yezoensis* farming activities usually occur between autumn and spring of the following year (starting in October, the net rope attached to the seaweed seedlings is placed on the farming raft, and harvested in stages from November to March of the following year). *Magallana gigas* farming continues throughout the year, and biomass varies with *M. gigas* harvest. We set up 15 sampling stations in the study area (Fig. 1), including 12 stations (S1-S12) in the *P. yezoensis*-*M. gigas* polyculture area and three stations (D1-D3) in the control area on the far shore. The distance of neighboring sites between culture and control area is approximately 3 km. Four field surveys were conducted in 2022–2023, with July, November, January and March representing summer, autumn, winter and spring, respectively.

2.2. Field sampling and analytical methods

In order to ensure the diversity of each site and the scientific validity of the data, all four surveys were conducted under relatively stable weather conditions with no wind or rain, and sample collection was completed between 8:00 and 16:00 on the same day. All samples were collected and stored strictly following the "Marine Monitoring Specifications" GB17378.3–2007. Surface water temperature (SST), salinity (S), pH (relative accuracy ± 0.01), and dissolved oxygen (DO) were measured at each sampling site using a multi-parameter water quality analyzer (YSI Pro Plus, USA). Each parameter was measured twice and averaged for analysis. Apparent oxygen consumption (AOU) is often used as a critical indicator of biological activity. It was calculated by subtracting the field-measured DO concentration from the air-equilibrated DO (Benson and Krause Jr, 1984). In this paper, AOU was used to assess the metabolic activities of organisms in different seasons

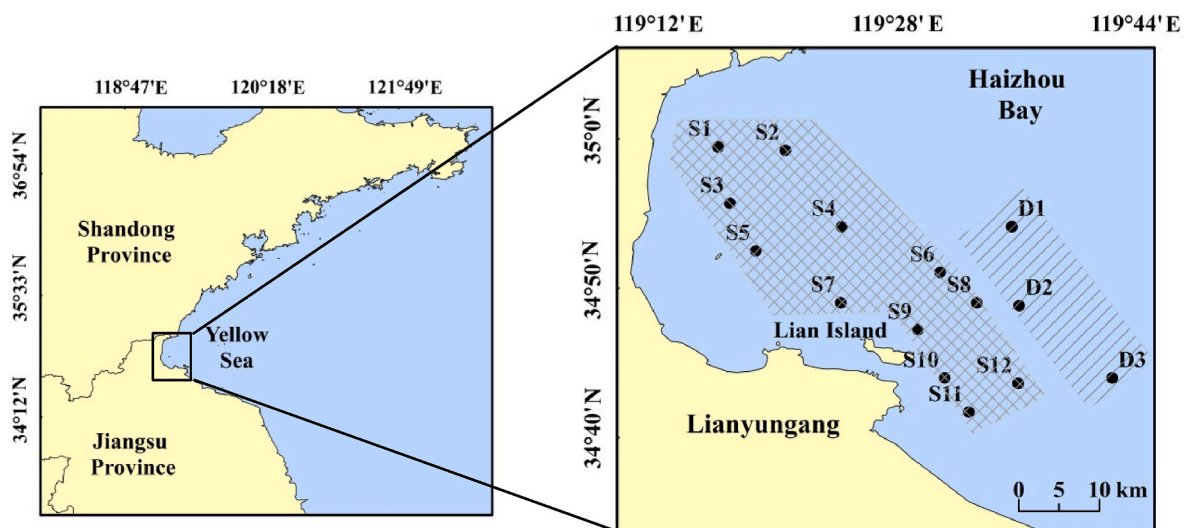


Fig. 1. Study site and the sampling stations in Haizhou Bay. S1-S12 are located in the *P. yezoensis*-*M. gigas* polyculture area and D1-D3 are in the control area.

without considering the sea-air exchange and water mixing effects; if $AOU < 0$, the marine ecosystem is dominated by photosynthesis; if $AOU > 0$, respiration is dominated (Zhang et al., 2023). Surface seawater samples (0.5 m below the surface) were collected using a stainless steel inverted water collector quickly introduced into polyethylene bottles, and stored under refrigeration at 4 °C for total alkalinity (TA) analyses; The water samples were filtered in the field using 0.45 µm cellulose ester microporous filter membranes under negative pressure, and the collected membranes and filtrate were frozen under light protection and transported to the laboratory within 24 h. The membranes were extracted with acetone (90%), and the Chla concentration was determined using a UV-visible spectrophotometer (UV-3200, MAPUDA, China). The filtrates were stored in glass bottles and kept in a refrigerator at -25 °C after transportation to the laboratory. The samples were removed and thawed at room temperature before analysis, and the DIC was measured in triplicate using a total organic carbon meter (TOC-L, SHIMADZU, Japan). Followed the national standard method in China, "The Specifications for Oceanographic Survey" (GB/T 12,763-2007), to the determination of TA in triplicate. Briefly, titrate a seawater sample with a standard hydrochloric acid solution to achieve a pH value between 3.40 and 3.90. Record the pH after titration and the amount of hydrochloric acid added. Use a 0.01 mol L⁻¹ sodium bicarbonate solution mixed with methyl red and methylene blue solutions as indicators to determine the concentration of the hydrochloric acid solution using titration.

The total alkalinity was calculated as follows:

$$A = \frac{V_{HCl} \times c(HCl)}{V_W} \times 1000 - \frac{\alpha_{H^+} \times (V_W + V_{HCl})}{V_W \times f_{H^+}} \quad (1)$$

Where A (µmol kg⁻¹) is the total alkalinity of seawater sample; V_{HCl} is the volume of standard hydrochloric acid solution, cm³; $c(HCl)$ (mol L⁻¹) is the concentration of standard hydrochloric acid solution; V_W (cm³) is the volume of seawater sample; α_{H^+} is the hydrogen ion activity corresponding to the pH value of the measured solution; f_{H^+} is the activity coefficient of hydrogen ions.

2.3. Calculation of the air-sea CO₂ fluxes

The air-sea CO₂ flux was estimated using the following formula:

$$F = k \times \alpha \times \Delta pCO_2 \quad (2)$$

Where F (mmol m⁻² d⁻¹) is the air-sea CO₂ fluxes, the positive or negative values indicate that the sea area is the source or sink of atmospheric CO₂, respectively; k (m s⁻¹) is the gas exchange coefficient and is calculated using the (Wanninkhof, 1992) model; k is calculated as follows:

$$K = 0.31 \times U_{10}^2 \times \left(\frac{Sc}{660} \right)^{-0.5} \quad (3)$$

Where U_{10}^2 (m s⁻¹) is the wind speed at 10 m above sea level, the wind speed data in this paper is provided by Ganyu Meteorological Station in Lianyungang, Jiangsu Province; Sc is the Schmidt coefficient, which is a function of temperature, $Sc = 2073.1 - 125.62t + 3.6276t^2 - 0.043219t^3$; α (mol kg⁻¹ atm⁻¹) is the solubility coefficient of CO₂ in seawater; ΔpCO_2 is the partial pressure of CO₂ (pCO_2) difference between surface seawater and the atmosphere. The pCO_2 of seawater is calculated using CO₂ SYS.XLS program based on data such as pH, TA, SST, and salinity (Prieto and Millero, 2002). In the subsequent discussion on the relationship between pCO_2 and environmental factors, in order to eliminate the effects of mixing and thermodynamic effects on pCO_2 , we normalized pCO_2 using CO₂SYS.XLS program. Use $npCO_{2@14.27^\circ C}$ to represent pCO_2 after normalization of average temperature and average salinity (Yang et al., 2021). In this study, we used 427.17 µatm as the mean atmospheric CO₂ concentration and the data were obtained from the

NOAA Global Monitoring Laboratory (<http://www.esrl.noaa.gov/gmd/ccgg/trends/global.html>).

2.4. Data analysis

Statistical analysis of the data was performed using SPSS 22.0 software. The effects of region and season on seawater carbonate system parameters and hydrological parameters were tested by analysis of variance (ANOVA). Use Ocean Data View software to draw contour maps. Perform linear regression analysis on the data using Graph Pad Prism 8.0.2 software to examine whether pCO_2 significantly correlates with different environmental factors and between each environmental factor. Principal component analysis was performed using Origin 2021 to identify the leading causes influencing the parameters of seawater carbonate systems. The significance level was set at $P < 0.05$ for all tests.

3. Results

3.1. Seasonal variations in water environmental parameters

The environmental and carbonate parameters of the surface seawater in the study area, according to the chronological order of the investigations are shown in Table 1. The annual variation of seawater temperature ranged from 5.1–28.4 °C, with highly significant differences between seasons ($P < 0.01$), with a decreasing trend in water temperature from the nearshore to the outside of the bay in spring, and an opposite trend in summer, autumn, and winter. In summary, pCO_2 showed a significant positive correlation with surface water temperature throughout the year (Fig. 2a). However, the correlation was not significant across seasons. The Salinity varied from 24.09 to 31.89 with significant seasonal and spatial variations ($P < 0.01$). The high rainfall and surface runoff in summer resulted in a significant decrease in salinity compared to other seasons. Regarding spatial variation, the salinity generally showed a gradual increase from near shore to outside the bay. From Fig. 2b, it can be seen that there is a significant negative correlation between pCO_2 and salinity throughout the year. However, in summer and autumn, pCO_2 is positively correlated with salinity due to freshwater input from rivers. While the correlation between pCO_2 and salinity is insignificant in winter and spring.

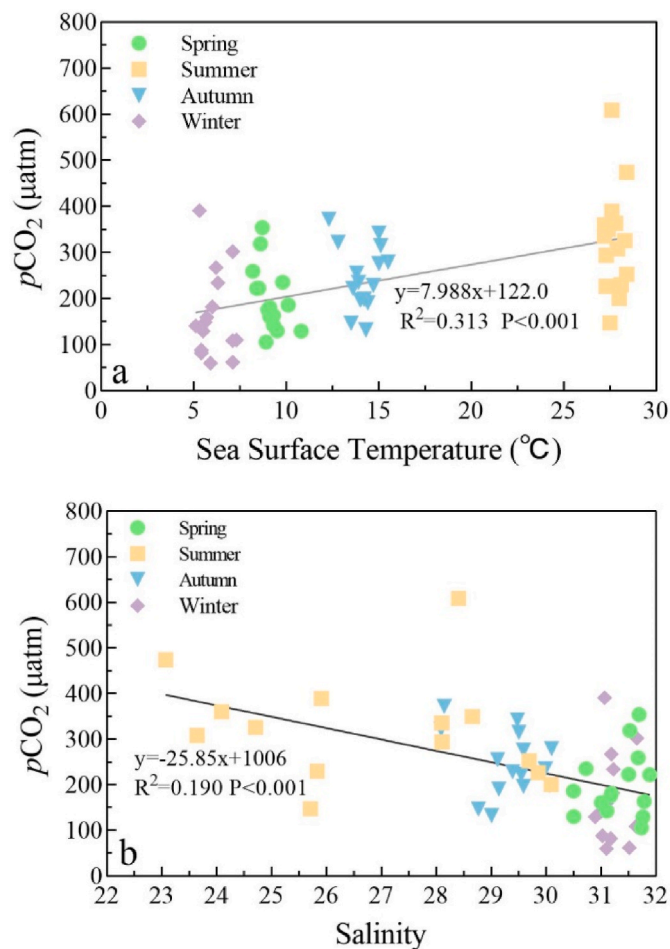
The variation range of DO concentration was 5.87–11.28 mg L⁻¹, with a very significant seasonal variation ($P < 0.01$), and the lowest value occurred in summer. The pH varied between 7.81–8.19 with highly significant seasonal and spatial variations ($P < 0.01$). The pH was lowest in summer and highest in winter, and the trend throughout the year increased from nearshore to outside the bay. The Chla concentrations ranged from 0.76–9.79 µg L⁻¹ with very significant seasonal variations ($P < 0.01$), and Chla concentrations were lowest in spring and highest in summer. There was a significant positive correlation between pH and DO concentrations throughout the year (Fig. 3a), where a significant correlation between pH and DO concentrations in the culture area was presented only in summer. Fig. 3b shows that there was no significant correlation between $npCO_{2@14.28^\circ C}$ and Chla concentration.

3.2. Seasonal variation of carbonate parameters and air-sea CO₂ fluxes

The variation of TA ranged from 2119.97 to 2440 µmol kg⁻¹ with significant differences between seasons ($P < 0.05$), whereas there was no significant difference between culture and control areas ($P > 0.05$). The high values of TA were mainly found in the culture areas in spring, autumn and winter, and the opposite was observed in summer. The variation range of DIC was between 1990.25–2437.15 µmol kg⁻¹, and the effects of season and region on DIC were insignificant ($P > 0.05$). The DIC concentrations in different seasons were in descending order: summer > autumn > winter > spring. There were significant differences between seasons in pCO_2 ($P < 0.01$), with the highest level (349.16 ± 114.90 µatm) occurring in the summer of monoculture *M. gigas* and the

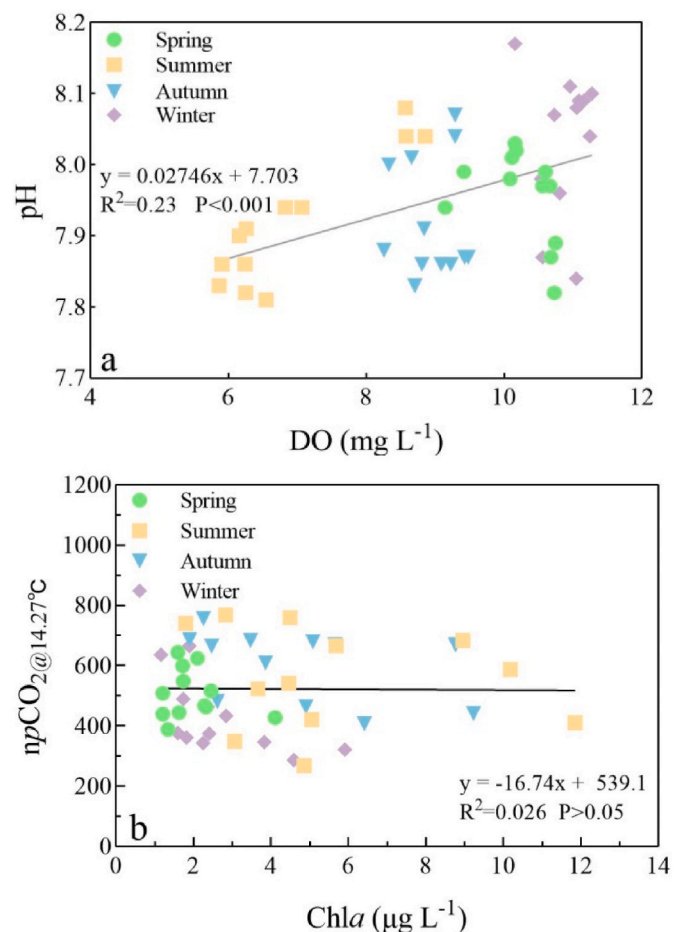
Table 1Seasonal variations of water environment parameters and carbonate parameters in the study area (Mean $\pm$ SD).

Season	Area	SST ($^{\circ}$ C)	S (‰)	DO (mg L $^{-1}$)	Chla (μ g L $^{-1}$)	pH	TA (μ mol kg $^{-1}$)	DIC (μ mol kg $^{-1}$)	pCO $_2$ (μ atm)
Summer	Pyropia-Oyster	27.69 $\pm$ 0.41	26.39 $\pm$ 1.99	6.93 $\pm$ 1.11	5.57 $\pm$ 2.79	7.66 $\pm$ 0.15	2298.56 $\pm$ 92.25	2253.25 $\pm$ 120.95	349.16 $\pm$ 114.90
	Control	27.90 $\pm$ 0.56	29.89 $\pm$ 0.20	8.37 $\pm$ 1.22	3.62 $\pm$ 0.63	7.92 $\pm$ 0.04	2306.95 $\pm$ 20.80	2226.15 $\pm$ 26.20	227.13 $\pm$ 25.87
Autumn	Pyropia-Oyster	13.96 $\pm$ 0.83	29.20 $\pm$ 0.63	8.96 $\pm$ 0.40	4.61 $\pm$ 2.22	7.92 $\pm$ 0.08	2401.54 $\pm$ 30.20	2263.68 $\pm$ 35.81	243.68 $\pm$ 74.41
	Control	14.87 $\pm$ 0.78	29.73 $\pm$ 0.32	8.94 $\pm$ 0.16	5.53 $\pm$ 0.63	7.98 $\pm$ 0.06	2364.63 $\pm$ 28.20	2198.63 $\pm$ 14.95	264.04 $\pm$ 60.49
Winter	Pyropia-Oyster	6.17 $\pm$ 0.81	31.23 $\pm$ 0.27	10.75 $\pm$ 0.44	3.22 $\pm$ 1.88	8.05 $\pm$ 0.10	2224.18 $\pm$ 96.24	2139.23 $\pm$ 107.84	161.99 $\pm$ 104.34
	Control	6.53 $\pm$ 0.68	31.37 $\pm$ 0.25	10.76 $\pm$ 0.33	2.23 $\pm$ 0.32	8.11 $\pm$ 0.04	2267.73 $\pm$ 64.57	2184.72 $\pm$ 57.74	175.52 $\pm$ 62.46
Spring	Pyropia-Oyster	9.15 $\pm$ 0.70	31.12 $\pm$ 0.58	10.18 $\pm$ 0.58	1.98 $\pm$ 0.84	7.96 $\pm$ 0.06	2250.13 $\pm$ 82.26	2148.05 $\pm$ 104.17	180.92 $\pm$ 67.48
	Control	8.73 $\pm$ 0.35	31.53 $\pm$ 0.46	10.04 $\pm$ 0.44	1.75 $\pm$ 0.38	8.03 $\pm$ 0.08	2270.65 $\pm$ 92.64	2163.17 $\pm$ 9.73	245.97 $\pm$ 98.86

**Fig. 2.** The relationship between $p\text{CO}_2$ and temperature (a), salinity (b) in surface seawater.

lowest level ($161.99 \pm 104 \mu\text{atm}$) occurring in the winter when the biomass of *P. yezoensis* was the largest. Fig. 4 shows the spatial distribution of $p\text{CO}_2$ and DIC in each season. In summer, $p\text{CO}_2$ showed a decreasing trend from nearshore to far shore. In contrast, the three seasons during the polyculture of *P. yezoensis* and *M. gigas* showed an opposite trend to summer, with $p\text{CO}_2$ showing a gradual increase from nearshore to far shore. Overall, $p\text{CO}_2$ in the study area shows an increasing trend with increasing seawater temperature. The distribution of DIC was similar to that of $p\text{CO}_2$, with generally higher levels of DIC in nearshore waters in all seasons.

The air-sea CO_2 flux showed significant seasonal differences ($P < 0.01$). The physiological activities of *M. gigas* and *P. yezoensis* have led to changes in the inorganic carbon system in the culture area, resulting in a very uneven distribution of air-sea CO_2 fluxes. Fig. 5 shows that the study area were the sink of atmospheric CO_2 during the culture period,

**Fig. 3.** The Relationship between pH and DO (a), $\text{npCO}_2@14.28^{\circ}\text{C}$ and Chla (b).

with average annual fluxes of $-26.32 \text{ mmol m}^{-2} \text{ d}^{-1}$ and $-27.93 \text{ mmol m}^{-2} \text{ d}^{-1}$ in the culture and control areas, respectively.

3.3. Factors affecting the seawater environment and the carbonate system

The AOU results of seawater in the culture area for each season were, in descending order, summer ($95.35 \mu\text{mol L}^{-1}$) > autumn ($44.01 \mu\text{mol L}^{-1}$) > spring ($16.47 \mu\text{mol L}^{-1}$) > winter ($-10.68 \mu\text{mol L}^{-1}$), in which the biological processes were dominated by photosynthesis in winter and respiration in other seasons. As shown in Fig. 6, there was a significant positive correlation between $\text{npCO}_2@14.28^{\circ}\text{C}$ and AOU throughout the year.

Principal component analysis (PCA) further revealed the impact of culture activities on the seawater environment. Since changes in DIC and TA can be significantly affected by salinity (Goyet et al., 1993), the parameters of the seawater carbonate system were normalized to mean salinity by the CO2SYS.XLS program before quantitative analysis using

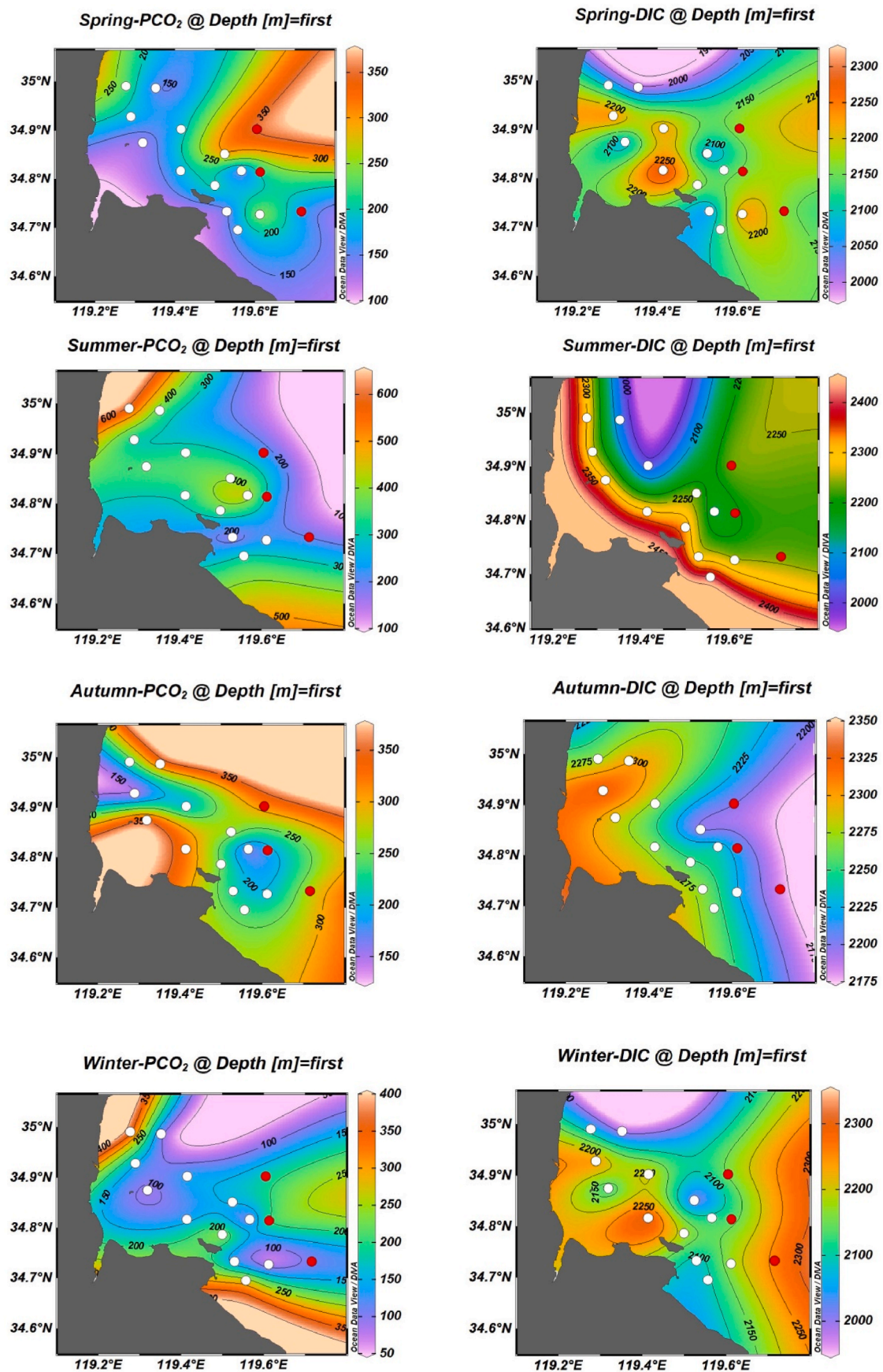


Fig. 4. Horizontal distribution of $p\text{CO}_2$ and DIC in different seasons (The white dots indicate the stations in the *P. yezoensis*-*M. gigas* polyculture area, and the red dots indicate the stations in the control area).

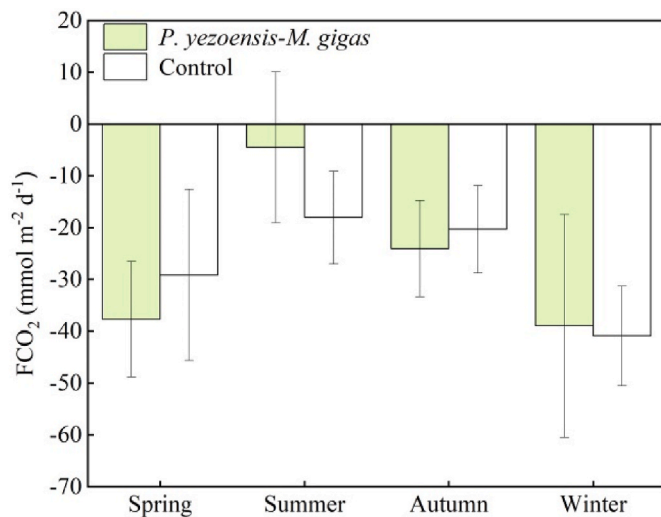


Fig. 5. Seasonal variation in air-sea CO₂ flux.

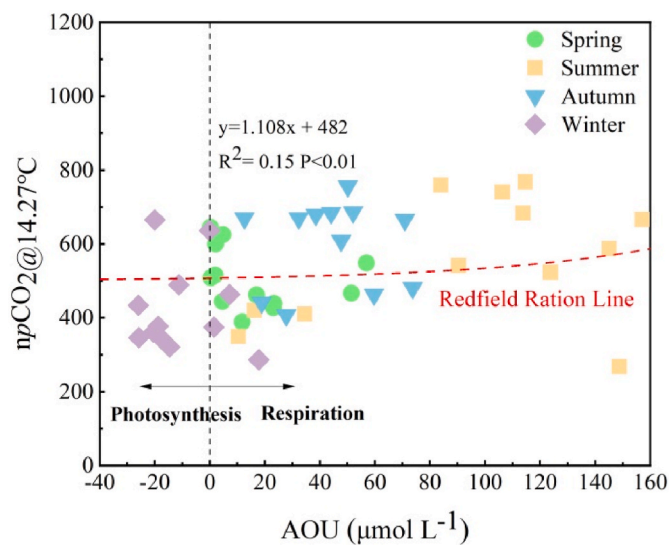


Fig. 6. The relationship between npCO₂@14.28°C and AOU.

the PCA model. The normalized parameters are nTA, nDIC, nHCO₃⁻, nCO₃²⁻ and nCO₂ (Millero et al., 1998). From the PCA model (Fig. 7), the first 2 principal components jointly reached 81.9%, and the first principal component contributed 50.3% of the variance. nTA, nDIC, nHCO₃⁻, and nCO₂ had a large loading on PC1, which reflected the direct effect of *M. gigas* culture on the carbonate system of seawater. pH and nCO₃²⁻ had a large positive loading on PC2 contributing 31.6% of the variance, and nCO₂ had a higher negative loading on PC2, which is hypothesized to be a possible combined effect of *P. yezoensis* and phytoplankton.

4. Discussion

4.1. The impact of water environmental factors on pCO₂

The pCO₂ in culture area is influenced by multiple factors such as water temperature, salinity (mixed with saltwater and freshwater), and the activities of mariculture organisms (Zhang et al., 2023). Surface water temperature is usually one of the main controlling factors for pCO₂. From the thermodynamic perspective of temperature in carbonate systems, pCO₂ increases with increasing temperature. Weiss. (1974) pointed out there is usually a significant positive correlation between

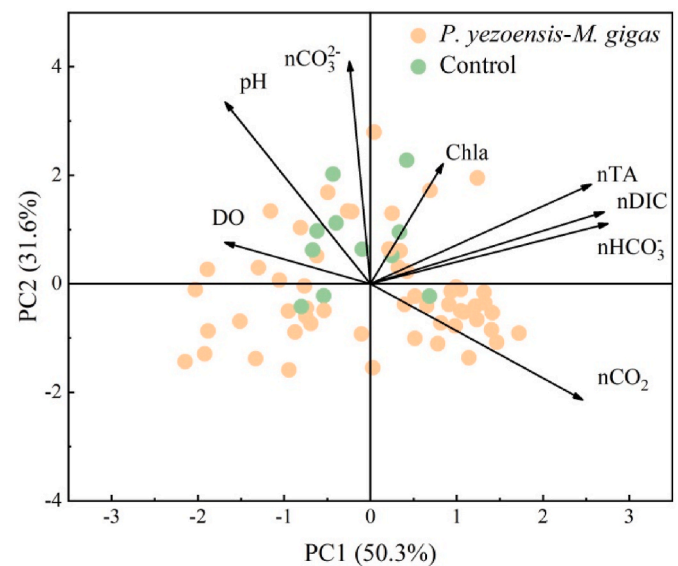


Fig. 7. Principal component analysis of water environment parameters and seawater carbonate system parameters.

pCO₂ and water temperature in temperate surface seawater. In order to determine whether water temperature is the main influencing factor of pCO₂ changes, we used the CO2SYS.XLS program to perform temperature compensation on pCO₂ in each season. After normalizing the temperature to the annual average temperature, we found that the trend and difference of changes between the culture and control areas remained consistent with the original results, the culture area > the control area. It can be inferred that temperature did not directly control the pCO₂ change. However, temperature was instead able to influence organisms' metabolic activities, thus indirectly controlling the change of pCO₂ (Chou et al., 2011). In addition, salinity is also one of the critical factors affecting pCO₂. Freshwater input can usually alter the concentrations of TA and DIC in seawater, thereby affecting the distribution of pCO₂ (Peng et al., 2006; Zhai et al., 2014). The linear relationship between pCO₂ and surface temperature and salinity indicated that both play a dominant role in the seasonal variation of pCO₂ in the study area. However, temperature and salinity are not the factors influencing the spatial distribution differences of pCO₂ in different seasons. According to the significant positive correlation between npCO₂@14.28°C and AOU, biological processes (photosynthesis and respiration) are also the main factors controlling the seasonal variation of pCO₂ in the study area.

In this study, *P. yezoensis* is less tolerant to high temperatures, so after its final harvest in the spring, only oyster culture activities were conducted in the culture area during the summer. During the summer, DO concentrations and seawater pH in the culture area were reduced compared to the control area, where the pH decreased by an average of 0.12 units. Wahl et al. (2018) shown that when the seawater pH decreased by 0.1 unit, the calcification rate of shellfish decreased by 13.5%. It is known that a decrease in seawater pH prevents calcifying organisms from precipitating calcium carbonate to form shells or skeletons. However, macroalgae farming provides a "mobile platform" to alleviate ocean acidification and protect vulnerable areas outside of wild macroalgae habitats through photosynthesis, as confirmed in this study (Xiao et al., 2021). We observed that during polyculture period, compared to summer, the pH value of seawater in the culture area significantly increased. At the same time, the difference in pH value between the culture area and the control area significantly decreased. This phenomenon suggests that *P. yezoensis* in the culture area provides a "shelter" for *M. gigas* to withstand the effects of ocean acidification. In addition, during the culture period, *P. yezoensis* can also absorb nitrogen released during *M. gigas* metabolism, thus effectively preventing the

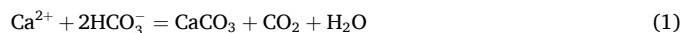
excessive depletion of nutrients in seawater (Chopin et al., 1999).

The suitable temperature conditions in summer and the lack of competition for nutrients from *P. yezoensis* promote the rapid growth of phytoplankton over a certain period. This can be confirmed by the observation that Chla concentrations in the study area were significantly higher in the summer than in other seasons and peaked in the culture area. In summer, there was a significant positive correlation between DO concentration and pH in the culture area, and the distribution pattern between $\text{npCO}_2@14.28^\circ\text{C}$ and AOU followed the trend of the Redfield Ration line. Among them, the AOU results of the three stations in the culture area were about $20\ \mu\text{mol L}^{-1}$, which was significantly lower than the results of the other stations. The pH values of these three stations (8.05 ± 0.02) exceeded the average summer level, and all of these results reflected the primary production process of phytoplankton. Therefore, when *P. yezoensis* does not provide "shelter" in summer, phytoplankton's photosynthesis may be the main reason for the pCO_2 desaturation of the culture area. However, throughout the observation period, phytoplankton growth was inhibited by *P. yezoensis* and the feeding or promoting effects of *M. gigas* on phytoplankton (Li et al., 2018; Peng et al., 2022). The activity of cultured organisms weakens the correlation between pCO_2 and Chla concentrations, resulting in no direct causal relationship between the two. During the polyculture period, the results of winter AOU reflected the photosynthetic oxygenation of *P. yezoensis*, and the DO of seawater reached a saturated state, which indicated that *P. yezoensis* culture was the main factor in reducing the pCO_2 of seawater in the culture area. The seawater in the culture area showed O_2 deficit in both autumn and spring. Photosynthesis of *P. yezoensis* was weak relative to *M. gigas* metabolic processes and oxygen-consuming decomposition processes of organic matter, which led to elevated AOU in the culture area.

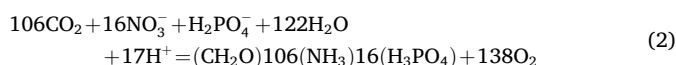
4.2. The impact of *P. yezoensis*-*M. gigas* polyculture on carbonate systems

The carbonate system controls seawater's pH and carbon flow between the biosphere, lithosphere, atmosphere and oceanosphere. However, Coastal ecosystems, especially when shallow and sheltered, are often subject to significant fluctuations in the carbonate system caused by hydrodynamics and biological activity (Müller et al., 2016). In recent years, *M. gigas* hanging cages have gradually been added around the *P. yezoensis* culture rafts in the study area. According to the mariculture data of Haizhou Bay in 2023, the culture area and annual production of *M. gigas* were $40\ \text{km}^2$ and $9 \times 10^4\ \text{t}$, respectively. Metabolic processes in shellfish can significantly influence seawater TA and DIC concentrations (Chauvaud et al., 2003). Therefore, large-scale *M. gigas* farming is one of the main drivers of changes in the parameters of the carbonate system in Haizhou Bay, thus changing the spatial variation of pCO_2 .

In summer (during oyster monoculture), high temperatures promote *M. gigas* respiration and bacterial decomposition of organic matter. Together with the release of CO_2 from mass-cultured *M. gigas* during the calcification process, these factors may elevate DIC concentrations and pCO_2 levels (Liu et al., 2017; Song et al., 2023). The ΔDIC between the culture and control areas was $82.33\ \mu\text{mol kg}^{-1}$, indicating that additional DIC was generated. However, one-way ANOVA showed no significant difference in DIC concentration between seasons ($P > 0.05$). This result may be because the calcification of *M. gigas* counteracts some of the DIC released by respiration. *Magallana gigas* are a type of shellfish with strong calcification. Ren et al. found through experiments that the calcification rate of mussel (*Mytilus galloprovincialis*) is $17.95 \pm 7.21\ \mu\text{mol g}^{-1}\ \text{h}^{-1}$, while the calcification rate of *M. gigas* can reach $56.37 \pm 14.85\ \mu\text{mol g}^{-1}\ \text{h}^{-1}$ (Ren, 2014). Studies have shown that DIC with reduced calcification in *M. gigas* can account for 61% of DIC with increased respiration (Ren et al., 2014). During the calcification process, the ratio of released CO_2 to precipitated CaCO_3 (ψ) fluctuates between 0.60 and 0.78 with the monthly water temperature variation (Han et al., 2021). The reaction equation for calcification is:



It can be seen that for every 1 mol of CaCO_3 precipitated, 2 mol of HCO_3^- will be absorbed. The CO_2 released as a by-product can be absorbed and utilized by phytoplankton and does not enter the atmosphere (Goll  ty et al., 2008; Su et al., 2020). On the other hand, the large amount of nutrients brought by river input can promote the primary production process of phytoplankton, thus consuming a large amount of DIC. In winter, the ΔDIC between the culture and control areas was $-62.30\ \mu\text{mol kg}^{-1}$. The salinity range of the study area was between 30.91–31.66, which was little affected by freshwater input, so the primary production process of *P. yezoensis* was the main reason for the DIC loss. Compared to DIC, TA concentrations in seawater were higher and less variable than DIC concentrations in all seasons. TA has good conserved properties, while DIC is susceptible to the comprehensive control of various factors such as the physiological activities of culture organisms, the input of terrestrial freshwater, and the exchange of air-sea interface (Goyet and Brewer, 1993; Tsunogai et al., 1999). The following equation can describe the photosynthesis of algae:



The primary production process reduces DIC concentrations and increases TA concentrations but does not significantly affect TA concentrations (Brewer and Goldman, 1976). However, in marine areas inhabited by many calcifying organisms, calcification can take up TA and DIC from seawater at a stoichiometric ratio of 2:1, which can have a significant effect on TA (Yang et al., 2023). Based on the 2:1 stoichiometric ratio, Li et al. estimated that *M. gigas* growth could reduce seawater TA concentration in Sanggou Bay by $52\ \mu\text{mol kg}^{-1}$ (Li et al., 2021). In this study, we observed a significant decrease in the TA concentration of seawater in the culture area in summer and spring (Fig. 8a). Among them, in spring, the maximum value of ΔTA was $156.74\ \mu\text{mol kg}^{-1}$ between the cultured area and the control area. The calcification of *M. gigas* was the main reason for the decrease in seawater TA concentration. Moreover, the prolongation of seawater residence time in the culture area also affected the changes in the carbonate system. The *M. gigas* culture method in Haizhou Bay is dominated by hanging cages, and the cages containing *M. gigas* can reach up to 3 m underwater. The densely distributed cages weakened the exchange capacity of seawater. They prolonged the residence time of seawater in the culture area, which made it easier to observe significant changes in the parameters of the seawater carbonate system. In addition, Aurora et al. found experimentally that the shell growth rate of juvenile *M. gigas* could increase by 40% compared to the control group with the increase of macroalgal biomass (Ricart et al., 2021). Therefore, it is also possible that the calcification of *M. gigas* may have been promoted with the increase in *P. yezoensis* biomass, leading to a significant decrease in TA concentration in the spring culture area.

In order to further study the impact of mariculture activities on the carbonate system in the study area, the ratio of DIC to TA (DIC: TA) was used to analyze the buffering capacity of the seawater carbonate system (Eggleston et al., 2010). The DIC: TA of the global ocean mixed layer is about 0.845–0.95, while the bays and adjacent seas affected by human activities tend to show a higher DIC: TA, such as the DIC: TA range of 0.92–0.99 in Hangzhou Bay in 2018 (Wang et al., 2021a). DIC: TA in seawater, on the one hand, depends on the characteristics of seawater's inorganic carbon chemical composition. On the other hand, it is related to the biogeochemical processes that dominate the region (Xu et al., 2023). It is generally believed that the closer DIC: TA is to 1, the weaker the buffering capacity of the seawater carbonate system. In shellfish farming areas, considering only shellfish respiration and metabolism, Yang et al. found that with the advancement of scallop culture, the DIC: TA in seawater increased from 0.89 to 0.92, and the buffering capacity of seawater showed a decreasing trend (Yang et al., 2021). In this study, as

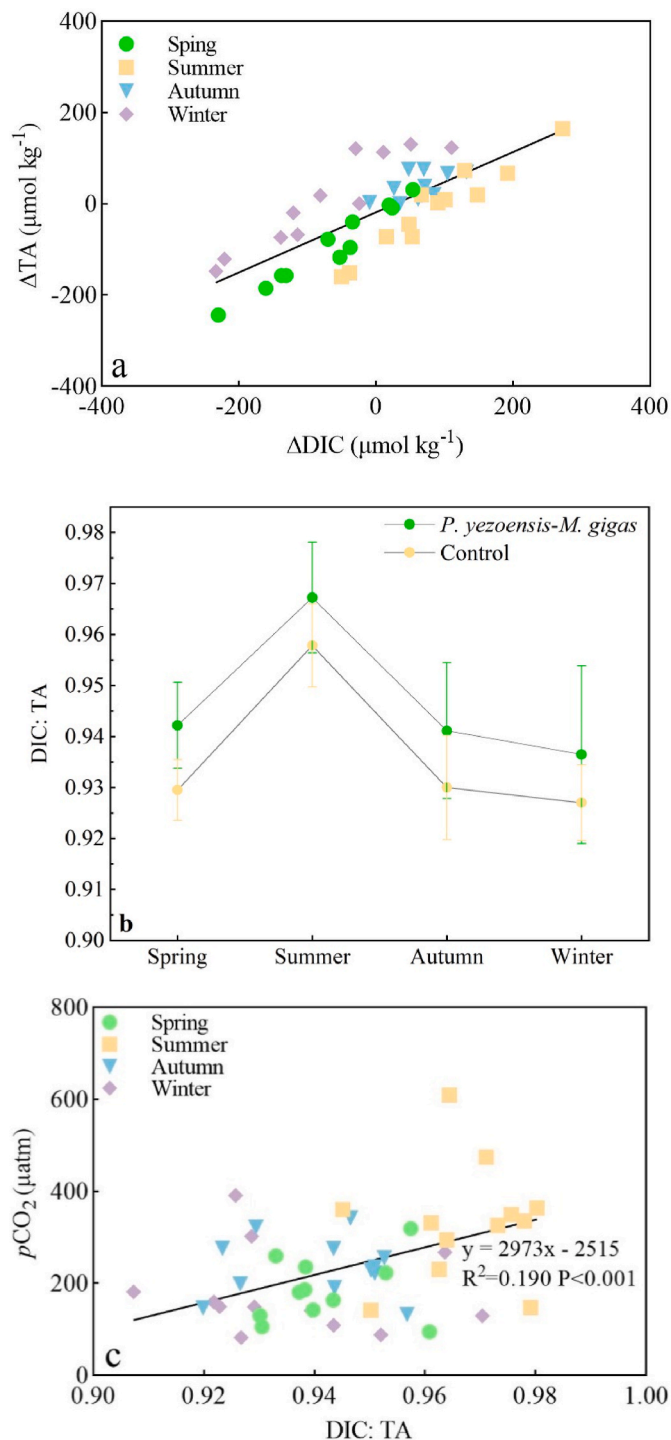


Fig. 8. Changes in TA and DIC differences between culture areas and control areas in different seasons (a), seasonal variation of DIC: TA (b), and the relationship between $p\text{CO}_2$ and DIC: TA (c).

shown in Fig. 8b, the seasonal changes of DIC: TA was summer > spring > autumn > winter in descending order. Especially in winter, when the growth of *P. yezoensis* is the most vigorous, the DIC: TA of seawater in the culture area showed a significant decreasing trend compared with that in summer, with its average value decreasing from 0.97 to 0.93. Moreover, there was a significant positive correlation between $p\text{CO}_2$ and DIC: TA (Fig. 8c). Based on the DIC and TA endmember values observed by Yu Siqing et al. during the 2018 Yellow Sea voyage, we calculated the DIC: TA of the water along the northern Jiangsu coast,

and the result was 0.94 (Yu, 2021). Based on this result, it can be inferred that summer *M. gigas* culture activities will reduce seawater's buffering capacity and accelerate seawater's acidification rate in the culture area.

4.3. Relationship between mariculture activities and air-sea CO_2 fluxes

The absorption capacity of CO_2 in the summer culture area was significantly weaker than in the *P. yezoensis-M. gigas* polyculture seasons but also showed a sink of atmospheric CO_2 , which was related to the contribution of phytoplankton. Han et al. (2021) showed that in spring, the Chl *a* concentrations in seawater from the shellfish culture areas of Shen'ao Bay and Sanggou Bay were $5.65 \pm 3.85 \mu\text{g L}^{-1}$ and $0.83 \pm 0.45 \mu\text{g L}^{-1}$, respectively. Among them, Shen'ao Bay exhibits a sink of atmospheric CO_2 , while Sanggou Bay exhibits a source of atmospheric CO_2 , which may be related to the higher primary productivity of Shen'ao Bay. Song et al. (2018) showed that the strong uptake of seawater CO_2 during the primary production of phytoplankton was the main reason for the lower than atmospheric level of $p\text{CO}_2$ at the sea surface of the South Yellow Sea in summer.

During the polyculture period, all three seasons showed atmospheric CO_2 sinks. The CO_2 uptake capacity was seasonally in descending order: winter, spring, and autumn, due to the differences in carbon sequestration rates of *P. yezoensis* at different growth stages. The lower seawater temperature in winter is favourable for the growth of *P. yezoensis*. As a result, the culture density and area of the *P. yezoensis* can reach the maximum in winter, and the absorption of CO_2 in the culture area was also the most significant (Wang et al., 2021b). Since spring was in the later stages of the *P. yezoensis* harvest, the biomass of *P. yezoensis* was reduced relative to winter. Therefore, the rate of CO_2 uptake in the culture area was lower than in winter. Compared to winter and spring, the rate of CO_2 uptake in the culture area was the smallest in autumn. *Pyropia yezoensis* in culture areas usually start releasing seedlings in October, and most of them were in a growing phase and less efficient in sequestering carbon during the autumn (November) survey cruises. Xue et al. (2019) also pointed out that when *P. yezoensis* was in the growth stage before harvesting, the absorption effect on CO_2 was significantly lower than that at the maturity stage. In addition, water temperatures are relatively high in autumn, and the physiological activity of *M. gigas* was generally consistent with changes in water temperature; for example, *M. gigas* have a higher respiration rate than in winter and spring (Lejart et al., 2012). Therefore, the seasonal characteristics of culture activities result in significantly weaker CO_2 absorption capacity in autumn culture areas compared to winter and spring.

5. Conclusion

Pyropia yezoensis-Magallana gigas polyculture activities are the main factors influencing significant seasonal changes in environmental factors and carbonate parameters of surface seawater in Haizhou Bay. Calcification of *M. gigas* significantly reduces the TA of seawater, thereby decreasing the CO_2 buffering capacity of seawater and making it a potential cause of ocean acidification. On the contrary, the photosynthesis of *P. yezoensis* and phytoplankton maintained a stable carbonate system. In summer, photosynthesis of phytoplankton and calcification of *M. gigas* consumed DIC in surface seawater, and the culture area showed a weak carbon sink. In the autumn, winter and spring, the culture area showed an obvious carbon sink because the *P. yezoensis-M. gigas* polyculture could effectively increase the rate of seawater CO_2 uptake. In conclusion, the polyculture model of *P. yezoensis-M. gigas* in Haizhou Bay effectively increased the carbon sink function of the coastal ecosystem while reducing the negative impacts of mariculture activities on the aquatic environment. This study mainly pointed out the effects of macroalgae-shellfish polyculture activities on the seawater environment, carbonate system, and the air-sea CO_2 flux. However, the carbon sequestration potential of cultured organisms should have been

evaluated in detail, and future research will focus on this.

CRedit authorship contribution statement

Jia-Hui Zhou: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Conceptualization, Formal analysis. **Ying-Zi Guo:** Investigation. **Xu Zhao:** Software, Resources, Investigation. **Jin-Chen Yu:** Writing – review & editing, Project administration, Investigation, Funding acquisition. **Jin Zhou:** Project administration, Investigation, Funding acquisition. **Jun Lin:** Resources. **Hong Huang:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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References

- Bartoli, M., Nizzoli, D., Viaroli, P., Turolla, E., Castaldelli, G., Fano, E.A., Rossi, R., 2001. Impact of Tapes philippinarum farming on nutrient dynamics and benthic respiration in the Sacca di Goro. *Hydrobiologia* 455, 203–212.
- Benson, B.B., Krause Jr., D., 1984. The concentration and isotopic fractionation of oxygen dissolved in freshwater and seawater in equilibrium with the atmosphere 1. *Limnol. Oceanogr.* 29, 620–632.
- Brewer, P.G., Goldman, J.C., 1976. Alkalinity changes generated by phytoplankton growth 1. *Limnol. Oceanogr.* 21, 108–117.
- Chauvaud, L., Thompson, J.K., Cloern, J.E., Thouzeau, G., 2003. Clams as CO₂ generators: the *Potamocorbula amurensis* example in San Francisco Bay. *Limnol. Oceanogr.* 48, 2086–2092.
- Chopin, T., Yarith, C., Wilkes, R., Belyea, E., Lu, S., Mathieson, A., 1999. Developing Porphyra/salmon integrated aquaculture for bioremediation and diversification of the aquaculture industry. *J. Appl. Phycol.* 11, 463–472.
- Chou, W., Gong, G., Tseng, C., Sheu, D., Hung, C., Chang, L., Wang, L., 2011. The carbonate system in the East China Sea in winter. *Mar. Chem.* 123, 44–55.
- Egleston, E.S., Sabine, C.L., Morel, F.M., 2010. Revelle revisited: buffer factors that quantify the response of ocean chemistry to changes in DIC and alkalinity. *Global Biogeochem. Cycles* 24.
- Fang, J., Fang, J., Chen, Q., Mao, Y., Jiang, Z., Du, M., Gao, Y., Lin, F., 2020. Assessing the effects of oyster/kelp weight ratio on water column properties: an experimental IMTA study at Sanggou Bay, China. *Journal of oceanology and limnology* 38, 1914–1924.
- Frankignoulle, M., Canon, C., Gattuso, J.P., 1994. Marine calcification as a source of carbon dioxide: positive feedback of increasing atmospheric CO₂. *Limnol. Oceanogr.* 39, 458–462.
- Gao, G., Beardall, J., Jin, P., Gao, L., Xie, S., Gao, K., 2022. A review of existing and potential blue carbon contributions to climate change mitigation in the Anthropocene. *J. Appl. Ecol.* 59, 1686–1699.
- Gao, G., Gao, L., Jiang, M., Jian, A., He, L., 2021. The potential of seaweed cultivation to achieve carbon neutrality and mitigate deoxygenation and eutrophication. *Environ. Res. Lett.* 17, 014018.
- Giles, H., Pilditch, C.A., 2004. Effects of diet on sinking rates and erosion thresholds of mussel *Perna canaliculus* biodeposits. *Mar. Ecol. Prog. Ser.* 282, 205–219.
- Golléty, C., Gentil, F., Davoult, D., 2008. Secondary production, calcification and CO₂ fluxes in the cirripedes *Chthamalus montagui* and *Elminius modestus*. *Oecologia* 155, 133–142.
- Gong, Q., 2022. Status survey and corresponding countermeasures of the laver industry development in lianyun District, Lianyungang city. In: Gong, Q. (Ed.). Yangzhou University.
- Goyet, C., Brewer, P.G., 1993. Biochemical Properties of the Oceanic Carbon Cycle, Modelling Oceanic Climate Interactions. Springer, pp. 271–297.
- Goyet, C., Millero, F.J., Poisson, A., Shafer, D.K., 1993. Temperature dependence of CO₂ fugacity in seawater. *Mar. Chem.* 44, 205–219.
- Han, T., Shi, R., Qi, Z., Huang, H., Gong, X., 2021. Impacts of large-scale aquaculture activities on the seawater carbonate system and air-sea CO₂ flux in a subtropical mariculture bay, southern China. *Aquaculture Environment Interactions* 13, 199–210.
- Han, T., Shi, R., Qi, Z., Huang, H., Liu, H., 2017. Interactive effects of oyster and seaweed on seawater dissolved inorganic carbon systems: implications for integrated multi-trophic aquaculture. *Aquaculture Environment Interactions* 9.
- Ikawa, H., Oechel, W.C., 2015. Temporal variations in air-sea CO₂ exchange near large kelp beds near San Diego, California. *J. Geophys. Res.: Oceans* 120 (1), 50–63.
- Jiang, Z., Li, J., Qiao, X., Wang, G., Bian, D., Jiang, X., Liu, Y., Huang, D., Wang, W., Fang, J., 2015. The budget of dissolved inorganic carbon in the shellfish and seaweed integrated mariculture area of Sanggou Bay, Shandong, China. *Aquaculture* 446, 167–174.
- Lejart, M., Clavier, J., Chauvaud, L., Hily, C., 2012. Respiration and calcification of *Crassostrea gigas*: contribution of an intertidal invasive species to coastal ecosystem CO₂ fluxes. *Estuar. Coast* 35, 622–632.
- Li, H., Zhang, Y., Liang, Y., Chen, J., Zhu, Y., Zhao, Y., Jiao, N., 2018. Impacts of maricultural activities on characteristics of dissolved organic carbon and nutrients in a typical raft-culture area of the Yellow Sea, North China. *Mar. Pollut. Bull.* 137, 456–464.
- Li, J., Zhang, W., Ding, J., Xue, S., Huo, E., Ma, Z., Yu, W., Jiang, Z., Fang, J., Mao, Y., 2021. Effect of large-scale kelp and bivalve farming on seawater carbonate system variations in the semi-enclosed Sanggou Bay. *Sci. Total Environ.* 753, 142065.
- Liu, Y., Lin, F., Wu, W., Wu, N., Zhang, Y., Wang, W., Li, M., Zhang, J., 2017. Seasonal variations (spring and summer) of the surface seawater pCO₂ in Sanggou Bay and corresponding impact factors. *J. Fish. Sci. China* 24, 1107–1114.
- Liu, Y., Zhang, J., Wu, W., Zhong, Y., Li, H., Wang, X., Yang, J., Zhang, Y., 2022. Effects of shellfish and macro-algae IMTA in north China on the environment, inorganic carbon system, organic carbon system, and sea–air CO₂ fluxes. *Front. Mar. Sci.* 9, 864306.
- Millero, F.J., Lee, K., Roche, M., 1998. Distribution of alkalinity in the surface waters of the major oceans. *Mar. Chem.* 60, 111–130.
- Müller, J.D., Schneider, B., Rehder, G., 2016. Long-term alkalinity trends in the Baltic Sea and their implications for CO₂-induced acidification. *Limnol. Oceanogr.* 61, 1984–2002.
- Pan, Z., Gao, Q., Dong, S., Wang, F., Li, H., Zhao, K., Jiang, X., 2019. Effects of abalone (*Haliotis discus hannai* Ino) and kelp (*Saccharina japonica*) mariculture on sources, distribution, and preservation of sedimentary organic carbon in Ailian Bay, China: identified by coupling stable isotopes ($\delta^{13}C$ and $\delta^{15}N$) with C/N ratio analyses. *Mar. Pollut. Bull.* 141, 387–397.
- Peng, P., Ma, Y., Shi, R., Wang, D., Xu, X., Yan, B., 2022. Temporal and spatial variation in the sea–air CO₂ flux in the oyster aquaculture area of Kozhou Bay. *Mar. Sci.* 46, 140–149.
- Peng, X., Ning, X., Sun, J., Le, F., 2006. Responses of phytoplankton growth on nutrient enrichments in the northern South China Sea. *Acta Ecol. Sin.* 26, 3959–3968.
- Prieto, F.J.M., Millero, F.J., 2002. The values of pK₁ + pK₂ for the dissociation of carbonic acid in seawater. *Geochem. Cosmochim. Acta* 66, 2529–2540.
- Qu, B., Song, J., Yuan, H., Li, X., Li, N., Duan, L., Chen, X., Lu, X., 2015. Summer carbonate chemistry dynamics in the southern Yellow Sea and the east China sea: regional variations and controls. *Continental Shelf Res.* 111, 250–261.
- Ren, L., 2014. In: Ren, L. (Ed.), Research on Carbon Sequestration of Cultured Oyster *Crassostrea gigas* and its Fouling Organisms in Sungen Bay. Doctoral Dissertation of Chinese Academy of Sciences, p. 130.
- Ren, L., Zhang, J., Fang, J., Yao, Y., Zhang, Y., Gao, Z., Zhang, M., 2014. Effects of calcification on respiratory quotient of cultured oyster *Crassostrea gigas* and its fouling animals. *J. Appl. Ecol.* 25, 1785–1790.
- Ricart, A.M., Gaylord, B., Hill, T.M., Sigwart, J.D., Shukla, P., Ward, M., Ninokawa, A., Sanford, E., 2021. Seagrass-driven changes in carbonate chemistry enhance oyster shell growth. *Oecologia* 196, 565–576.
- Sabine, C.L., Feely, R.A., Gruber, N., Key, R.M., Lee, K., Bullister, J.L., Wanninkhof, R., Wong, C., Wallace, D.W., Tilbrook, B., 2004. The oceanic sink for anthropogenic CO₂. *Science* 305, 367–371.
- Shadwick, E.H., Thomas, H., Azetsu-Scott, K., Greenan, B.J., Head, E., Horne, E., 2011. Seasonal variability of dissolved inorganic carbon and surface water pCO₂ in the Scotian Shelf region of the Northwestern Atlantic. *Mar. Chem.* 124, 23–37.
- Song, C., Xiong, Y., Jin, P., Sun, Y., Zhang, Q., Ma, Z., Gao, G., 2023. Mariculture structure adjustment to achieve China's carbon neutrality and mitigate climate change. *Sci. Total Environ.* 895, 164986.
- Song, J., Qu, B., Li, X., Yuan, H., Li, N., Duan, L., 2018. Carbon sinks/sources in the Yellow and East China Seas—air-sea interface exchange, dissolution in seawater, and burial in sediments. In: Song, J., Qu, B., Li, X., Yuan, H., Li, N., Duan, L. (Eds.), *Science China Earth Sciences*, pp. 1583–1593.
- Su, J., Cai, W.-J., Brodeur, J., Chen, B., Hussain, N., Yao, Y., Ni, C., Testa, J.M., Li, M., Xie, X., 2020. Chesapeake Bay acidification buffered by spatially decoupled carbonate mineral cycling. *Nat. Geosci.* 13, 441–447.
- Tang, Q., Zhang, J., Fang, J., 2011. Shellfish and seaweed mariculture increase atmospheric CO₂ absorption by coastal ecosystems. *Mar. Ecol. Prog. Ser.* 424, 97–104.
- Tsunogai, S., Watanabe, S., Sato, T., 1999. Is there a "continental shelf pump" for the absorption of atmospheric CO₂. *Tellus B* 51.
- Wahl, M., Schneider Covachá, S., Saderne, V., Hiebenthal, C., Müller, J., Pansch, C., Sawall, Y., 2018. Macroalgae may mitigate ocean acidification effects on mussel calcification by increasing pH and its fluctuations. *Limnol. Oceanogr.* 63, 3–21.
- Wang, J., Wang, B., Li, D., 2021a. Characteristics of carbonate system in the Hangzhou Bay: under the regulation of air-sea exchange and respiration. *Hai Yang Xue Bao* 43, 21–32.
- Wang, X., Liu, J., Xing, Q., Chen, Y., 2021b. Monitoring of Porphyra cultivation dynamics in Lianyungang based on coastal zone imager. *Mar. Sci.* 45, 9–17.

- Wanninkhof, R., 1992. Relationship between wind speed and gas exchange over the ocean. *J. Geophys. Res.: Oceans* 97, 7373–7382.
- Weiss, R.F., 1974. Carbon dioxide in water and seawater: the solubility of a non-ideal gas. *Mar. Chem.* 2, 203–215.
- Wu, X., Yuan, X., Liu, X., Wang, X., Chen, Z., Gu, T., Zhang, G., Gong, X., Sun, J., 2024. Distributions and controlling processes of the carbonate system in the Eastern Indian Ocean during autumn and spring. *Front. Mar. Sci.* 11, 1343690.
- Xiao, X., Agustí, S., Yu, Y., Huang, Y., Chen, W., Hu, J., Li, C., Li, K., Wei, F., Lu, Y., 2021. Seaweed farms provide refugia from ocean acidification. *Sci. Total Environ.* 776, 145192.
- Xu, X., Wang, X., Hu, H., Xiang, R., Cai, Y., Liu, C., Zhang, W., Chen, J., Ma, Y., 2023. Hourly variations of partial pressure of CO₂ in surface sea water and its controlling mechanisms in the northeastern Beibu Gulf in spring and summer. *Hai Yang Xue Bao* 45, 14–26.
- Xue, F., Peng, X., Ke, A., 2019. CO₂ footprints and fluxes at the air-sea interface during fall and winter in seawater culture areas of Porphyra. *Zhejiang Agricultural Science* 60, 1029–1035.
- Yan, L., Huang, H., Chen, J., Yang, X., 2011. Estimation of carbon sink capacity of algal mariculture in the coastal areas of China. *Adv. Mar. Sci.* 29, 537–545.
- Yang, B., Gao, X., Zhao, J., Liu, Y., Lui, H., Huang, T., Chen, C., Xing, Q., 2021. Massive shellfish farming might accelerate coastal acidification: a case study on carbonate system dynamics in a bay scallop (*Argopecten irradians*) farming area, North Yellow Sea. *Sci. Total Environ.* 798, 149214.
- Yang, B., Zhang, Z., Cui, Z., Xie, Z., Chen, B., Zheng, H., Liao, B., Zhou, J., Xiao, B., 2023. Multiple factors driving carbonate system in subtropical coral community environments along Dapeng Peninsula, South China Sea. *Atmosphere* 14, 688.
- Yool, A., Fasham, M.J.R., 2001. An Examination of the “Continental shelf pump” in an open ocean general circulation model. *Global Biogeochem. Cycles* 15, 831–844.
- Yu, S., 2021. Quasi-synchronous Cumulative Oxygen Utilization and Dissolved Inorganic Carbon in the South Yellow Sea Cold Water Mass: Regulation of Community Metabolism and Hydrological Processes. ShanDong University.
- Zhai, W.D., Chen, J.F., Jin, H.Y., Li, H.L., Liu, J.W., He, X.Q., Bai, Y., 2014. Spring carbonate chemistry dynamics of surface waters in the northern East China Sea: water mixing, biological uptake of CO₂, and chemical buffering capacity. *J. Geophys. Res.: Oceans* 119, 5638–5653.
- Zhang, J., Wu, W., Ren, L., Han, T., Wang, W., Fang, J., 2013. Seasonal variation of pCO₂ and its potential influencing factors in aquaculture areas of Sanggou Bay. *Progress in Fishery Sciences* 34, 57–64.
- Zhang, Y., Zhang, J., Liang, Y., Li, H., Li, G., Chen, X., Zhao, P., Jiang, Z., Zou, D., Liu, X., 2017. Carbon sequestration processes and mechanisms in coastal mariculture environments in China. *Sci. China Earth Sci.* 60, 2097–2107.
- Zhang, Z., Yang, B., Cui, Z., Lin, Z., Xie, Z., 2023. Distribution characteristics of pCO₂ and source-sink conversion mechanism of CO₂ in a subtropical coral reef water during the dry season: a case study of the Yangmeikeng Sea area in Shenzhen, China. *Mar. Sci.* 47, 45–56.
- Zhou, Y., Chen, Y., Yang, J., Wang, J., Lu, M., 2024. Assessment and analysis of water quality in main estuaries of Haizhou bay. *Leather making and environmental technology* 5, 167–169.